

Optimal lattices for a three-body power-law energy: steep-decay limits, computer-assisted proofs and the minima of the modular graph functions $C_{s,s,s}$

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Abstract We study which Bravais lattice of fixed density minimises the three-body energy $T_v(\Lambda) = \sum'_{x,y \in \Lambda} (|x||y||x-y|)^{-v}$, the sum over all lattice triangles through the origin of the product of their side lengths to the power $-v$. In two dimensions T_{2s} coincides, up to a factor, with the two-loop modular graph function $C_{s,s,s}(\tau)$. We prove that in the steep-decay limit $v \rightarrow \infty$ the minimisers converge to the rectangular lattice with aspect ratio $\sqrt{(\sqrt{17}-1)/2}$ in $d=2$, by solving exactly the associated max–min problem for the smallest triangle product, and to the body-centred cubic lattice in $d=3$, by a computer-assisted argument. For finite exponents we give computer-assisted proofs, carried out in rigorous interval arithmetic, that $C_{2,2,2}$, $C_{3,3,3}$ and $C_{4,4,4}$ attain their minimum only at the square point $\tau = i$ (whereas $C_{1,1,1}$ is minimal at the hexagonal point), that the square lattice is the unique minimiser also for $v=5,7$, that the hexagonal lattice is the unique minimiser for $v=3.5$, that the hexagonal and square energies cross in $(3.918364, 3.918366)$, and that the square lattice loses local minimality at some $v_2 \in (8.604, 8.608)$. Beyond the limits, we show that for all sufficiently large exponents the minimiser is *exactly* BCC in $d=3$. In $d=2$ it is the rectangular lattice whose aspect ratio solves $(y\sqrt{1+y^2}/2)^v = 4(y^2-1)/(3(y^2+1))$ up to an exponentially small error, so that $y_v = y_\infty - \kappa/v + O(v^{-2})$ with an explicit $\kappa \approx 0.95489$. This rests on a local uniqueness principle for smoothed max–min problems, a many-body version of Ryshkov’s argument for the Epstein zeta function. The computer-assisted proofs rest on a majorant series that bounds all Taylor coefficients of every summand uniformly in the lattice vectors; the elementary planar results are also formalised in Lean 4. Numerically, the planar minimiser switches discontinuously from hexagonal to square at $v^* \approx 3.91836$ and bifurcates continuously from square to rectangular at v_2 . In three dimensions BCC is the best lattice found for all tested exponents $3.2 \leq v \leq 12$, FCC loses local stability at $v_c = 3.7521 \pm 0.0001$, and hcp with $c/a \in [1.45, 1.85]$ is never optimal at the tested exponents.

Keywords Lattice energy minimisation · Three-body interactions · Modular graph functions · Computer-assisted proof · Max–min problems · Crystallisation

1 Introduction

Which periodic arrangement of points minimises a given interaction energy is one of the classical questions linking number theory, geometry and the physics of crystals. For *pair* interactions a great deal is known: for completely mono-

tone functions of the squared distance the hexagonal lattice is optimal in the plane among all lattices of fixed density [1, 2, 3, 4, 5], universal optimality has been established for E_8 and the Leech lattice [6], and phase transitions between optimal lattices have been classified for many two-term and ratio-type pair energies [7, 8, 9]. In three dimensions the optimality of the face-centred cubic lattice for Epstein zeta functions (the Sarnak–Strömbergsson conjecture) has recently been claimed [10]. We refer to [18] for a survey of the crystallisation problem.

Much less is known for genuine *many-body* energies. The simplest one is the product-form three-body power law

$$T_v(\Lambda) = \sum_{\substack{x,y \in \Lambda \setminus \{0\} \\ x \neq y}} (|x||y||x-y|)^{-v}, \quad (1)$$

which converges for $v > 2d/3$. At $v=3$ it is the radial part of the Axilrod–Teller–Muto (ATM) triple-dipole energy; under the name *three-body zeta function* it was introduced and evaluated for selected lattices and along the Bain path in [11], and efficient algorithms for such many-body lattice sums were developed in [12, 13]. We stress that (1) is a model energy: physical three-body dispersion carries an angular factor and always comes with pair terms. Three-body terms do appear in rigorous crystallisation results for finite particle systems, but in a different form: as *angular* potentials that favour prescribed bond angles, in the square lattice [19, 20], in general stratified settings [21, 22] and for the FCC lattice [23]; many-body embedded-atom models were treated in [24]. These works minimise over configurations with short-range interactions and do not concern the long-range power-law sum (1) over lattices. Our motivation is mathematical. In $d=2$, the energy T_{2s} is (up to normalisation) the two-loop *modular graph function* $C_{s,s,s}(\tau)$ from the low-energy expansion of genus-one string amplitudes [15]; its minimum over the fundamental domain appears not to have been studied.

Two remarks delimit the content. First, T_v is invariant under isometries, so the hexagonal and the square lattice (the elliptic fixed points $\rho = e^{i\pi/3}$ and i of the modular group) are always critical points [16]. The substance of the results below is *global* minimality and the location of the switches between minimisers. Second, the reduction of a steep-decay limit to a packing-type problem is standard for pair energies [7]; here it leads to a new max–min problem for triangles, related in spirit to three-point packing problems on the sphere [17].

Summary of results. Fig. 1 summarises the planar picture.

- Theorem 1 solves exactly the planar max–min problem for the smallest triangle product; Theorem 2 deduces that minimisers of T_V converge to a rectangular lattice with aspect ratio $y_\infty = \sqrt{(\sqrt{17}-1)/2}$.
- Theorem 3 (computer-assisted) shows that in $d = 3$ the body-centred cubic (BCC) lattice is the unique maximiser of the smallest triangle product; minimisers of T_V converge to BCC.
- Theorems 4 and 5 go beyond the limits. For all sufficiently large v the minimiser is exactly BCC in $d = 3$. In $d = 2$ it is a rectangular lattice whose aspect ratio is given, up to an exponentially small error, by an explicit transcendental equation. Both follow from a general local uniqueness principle (Lemma 7).
- Theorem 6 (computer-assisted, rigorous interval arithmetic) establishes the minimiser for the finite exponents $v \in \{3.5, 4, 5, 6, 7, 8\}$; in particular $C_{2,2,2}$, $C_{3,3,3}$, $C_{4,4,4}$ are minimal only at $\tau = i$. It also encloses the crossing v^* and the loss of stability v_2 of the square lattice.
- Sections 7 and 8 give the numerical phase diagrams with error bars, including hcp, unequal exponents and mixed pair plus three-body energies.

2 Setting and the modular graph function

Throughout, lattices are Bravais lattices of unit covolume, considered up to isometry. For three pairwise distinct points p, q, r put $\pi(p, q, r) = |p - q||q - r||r - p|$, and let

$$P(\Lambda) = \min\{\pi(p, q, r) : p, q, r \in \Lambda \text{ pairwise distinct}\},$$

the smallest triangle product; collinear triples are admitted. Every triple can be translated to contain the origin, hence

$$T_V(\Lambda) \geq P(\Lambda)^{-v}. \quad (2)$$

In $d = 2$ we write $\Lambda_\tau = (\mathbb{Z} + \tau\mathbb{Z})/\sqrt{\text{Im } \tau}$ with $\tau = x + iy$ in the fundamental domain. The modular graph function of [15] is $C_{a,b,c}(\tau) = \sum_{p_1+p_2+p_3=0, p_i \neq 0} \prod_i (\text{Im } \tau / \pi |p_i|^2)^{a_i}$, $p_i \in \mathbb{Z} + \tau\mathbb{Z}$, and a direct comparison gives

$$C_{s,s,s}(\tau) = \pi^{-3s} T_{2s}(\Lambda_\tau). \quad (3)$$

At $s = 1$, Zagier's identity $C_{1,1,1} = E_3 + \zeta(3)$ and the minimality of the Epstein zeta function at ρ [1, 2, 3, 4] show that $C_{1,1,1}$ is minimal only at ρ . We verified (3) numerically, and the identities $C_{1,1,1} = E_3 + \zeta(3)$ and $C_{2,2,1} = \frac{2}{5}E_5 + \zeta(5)/30$ to relative accuracy 10^{-14} and 10^{-15} at five values of τ (Appendix A).

3 The planar steep-decay limit

Let $a_* = ((1 + \sqrt{17})/8)^{1/4} \approx 0.894563$, so that $4a_*^8 = a_*^4 + 1$, and put

$$P_* = 2a_*^3 = \sqrt{a_*^2 + a_*^{-2}} \approx 1.431734, \quad \Lambda_* = a_*\mathbb{Z} \times a_*^{-1}\mathbb{Z}.$$

The aspect ratio of Λ_* is $y_\infty = a_*^{-2} = \sqrt{(\sqrt{17}-1)/2} \approx 1.249621$.

Theorem 1. For every lattice $\Lambda \subset \mathbb{R}^2$ of unit covolume, $P(\Lambda) \leq P_*$, with equality if and only if Λ is isometric to Λ_* .

Lemma 1. After a rotation and a reflection, Λ has a basis $u = (a, 0)$, $v = (p, a^{-1})$ with $a = \lambda_1(\Lambda)$, $0 \leq p \leq a/2$ and $|v| \geq a$; moreover $a^4 \leq 4/3$.

Proof. Lagrange–Gauss reduction; the bound is Hermite's constant $\gamma_2 = 2/\sqrt{3}$. $\square$

Lemma 2. In the notation of Lemma 1, $P(\Lambda) \leq G(a) := \min\{2a^3, \sqrt{a^2 + a^{-2}}\}$, and $\pi(0, u, v) \leq \sqrt{a^2 + a^{-2}}$ with equality if and only if $p = 0$.

Proof. The collinear triple $(0, u, 2u)$ gives $2a^3$. For $(0, u, v)$ we have $\pi^2 = a^2(p^2 + k)((a-p)^2 + k)$ with $k = a^{-2}$. With $m = p(a-p) \in [0, a^2/4]$ and $p^2 + (a-p)^2 = a^2 - 2m$,

$$(p^2 + k)((a-p)^2 + k) = (ka^2 + k^2) + m(m-2k).$$

Since $a^4 < 8$, $m \leq a^2/4 < 2k$, so the last term is ≤ 0 , with equality iff $m = 0$, i.e. $p = 0$. Hence $\pi^2 \leq a^2(ka^2 + k^2) = a^2 + a^{-2}$. $\square$

Lemma 3. For $0 < a \leq (4/3)^{1/4}$, $G(a) \leq P_*$ with equality iff $a = a_*$.

Proof. For $a \leq a_*$, $G(a) \leq 2a^3 \leq 2a_*^3$. The function $g(a) = a^2 + a^{-2}$ decreases on $(0, 1]$ and increases on $[1, \infty)$. For $a_* < a \leq 1$, $G(a) \leq \sqrt{g(a)} < \sqrt{g(a_*)} = P_*$. For $1 \leq a \leq (4/3)^{1/4}$, $g(a) \leq \frac{2}{\sqrt{3}} + \frac{\sqrt{3}}{2} = \frac{7}{2\sqrt{3}}$ and $(\frac{7}{2\sqrt{3}})^2 = \frac{49}{12} < 16(\frac{1+\sqrt{17}}{8})^3 = P_*^4$. $\square$

Lemma 4. $P(\Lambda_*) = P_*$.

Proof. The triples in Fig. 1(a) give $P(\Lambda_*) \leq P_*$. Collinear triples lie on a lattice line with spacing $s \geq a_*$ and have $\pi = s^3(k-j)(l-k)(l-j) \geq 2s^3 \geq P_*$. A non-collinear lattice triangle has area $\geq \frac{1}{2}$; with side lengths $d_1 \leq d_2 \leq d_3$ and γ the angle between the two shorter sides, $d_1 d_2 \geq d_1 d_2 \sin \gamma \geq 1$, hence $\pi \geq d_3$. The only vectors of Λ_* shorter than $D = \sqrt{a_*^2 + a_*^{-2}}$ are $\pm u, \pm v$ with $u = (a_*, 0)$, $v = (0, a_*^{-1})$: for (ma_*, na_*^{-1}) with $n \neq 0$ one gets $m = 0$ and $n^2 < 1 + a_*^4 < 2$, and for $n = 0$, $m^2 < 1 + y_\infty^2 < 4$. They span two directions, whereas the three sides of a triangle are pairwise linearly independent. Thus $d_3 \geq D = P_*$. $\square$

Proof of Theorem 1. Lemmas 1–3 give $P(\Lambda) \leq G(a) \leq P_*$. Equality forces $a = a_*$ and $\pi(0, u, v) \geq P_* = \sqrt{a_*^2 + a_*^{-2}}$, hence $p = 0$ by Lemma 2, i.e. $\Lambda = \Lambda_*$. The converse is Lemma 4. $\square$

Theorem 2. For every $v > 4/3$, T_V attains its minimum among planar lattices of unit covolume. If Λ_v is any minimiser, then $\Lambda_v \rightarrow \Lambda_*$ up to isometries as $v \rightarrow \infty$, and $(\min T_V)^{1/v} \rightarrow 1/P_*$. Equivalently, minimisers τ_s of $C_{s,s,s}$ on the fundamental domain satisfy $\tau_s \rightarrow iy_\infty$ as $s \rightarrow \infty$.

Proof. P is continuous and $P(\Lambda) \leq 2\lambda_1(\Lambda)^3$; T_V is lower semicontinuous as a sum of non-negative continuous functions. By (2), sublevel sets of T_V lie in $\{\lambda_1 \geq \varepsilon\}$, compact by Mahler's criterion, so minimisers exist. Every term of

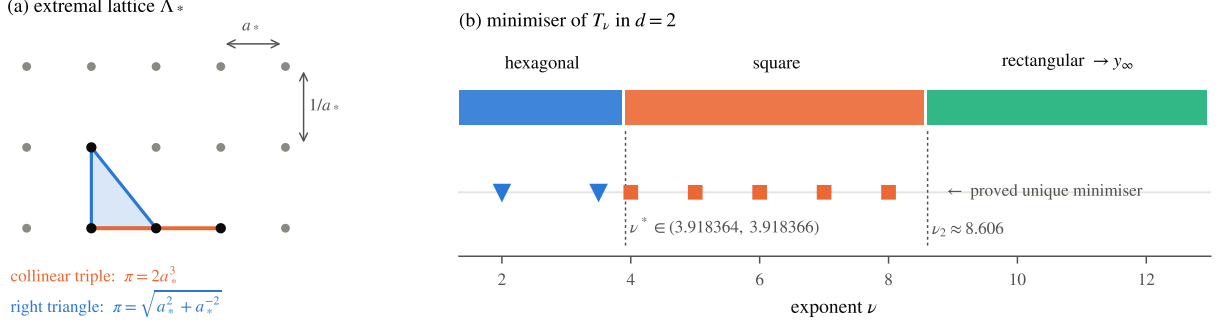


Fig. 1 (a) The extremal lattice Λ_* of Theorem 1: the collinear triple (red) and the right triangle (blue) have the same side-length product P_* . (b) Minimiser of T_ν in $d=2$. Shaded regions: numerical phase diagram (Sect. 7); symbols: exponents at which the minimiser is established rigorously ($\nu=2$ classical, the others by Theorem 6). For $\nu \rightarrow \infty$ the minimiser converges to Λ_* (Theorem 2).

$T_\nu(\Lambda_*)$ has $\pi \geq P_*$, whence $T_\nu(\Lambda_*) \leq CP_*^{-\nu}$ for $\nu \geq \nu_0$ with $C = T_{\nu_0}(\Lambda_*)P_*^{\nu_0}$. With (2), $P(\Lambda_\nu)^{-\nu} \leq T_\nu(\Lambda_\nu) \leq T_\nu(\Lambda_*) \leq CP_*^{-\nu}$, so $P_* \geq P(\Lambda_\nu) \geq C^{-1/\nu}P_* \rightarrow P_*$. The Λ_ν lie in a compact set; every limit point has $P = P_*$ and is Λ_* by Theorem 1. The statement for $C_{s,s,s}$ follows from (3). $\square$

The numerical minimisers for finite ν (Sect. 7) approach Λ_* monotonically: the optimal aspect ratio is 1.056, 1.100, 1.141, 1.170, 1.194 and 1.215 at $\nu = 9, 10, 12, 15, 20, 30$. A random test on $2 \cdot 10^4$ lattices confirmed the bound of Lemma 2, and a grid maximisation of P located the maximum at $\tau \approx 1.2475i$.

Formal verification. Both mathematical parts of the proof of Theorem 1 are formalised in Lean 4 with Mathlib. The first is the upper bound $\min\{4a^6, \Phi(a, p)\} \leq P_*^2$ for reduced bases, with $\Phi(a, p) = \pi(0, u, v)^2$, together with its equality case $a = a_*$, $p = 0$ (Lemmas 2 and 3). The second is the lower bound for all lattice triangles of the rectangular lattices $\sqrt{s}\mathbb{Z} \times \mathbb{Z}/\sqrt{s}$ with $1/3 < s^2 < 1$ (Lemma 4). The formalisation contains no sorry, and #print axioms reports only the standard axioms (propxt, Classical.choice, Quot.sound). Not formalised are the classical facts used in Lemma 1 (existence of a reduced basis, Hermite's bound), the translation invariance of π , and the compactness argument of Theorem 2.

4 Three dimensions

In $d=3$ the energy converges for $\nu > 2$. For BCC of unit covolume $P(\text{BCC}) = 3/2$, attained by isosceles triangles with two nearest-neighbour sides and one second-neighbour side; in contrast $P(\text{FCC}) = P(\text{SC}) = P(\text{hcp}) = \sqrt{2}$.

Theorem 3 (computer-assisted). *For every lattice $\Lambda \subset \mathbb{R}^3$ of unit covolume, $P(\Lambda) \leq 3/2$, with equality iff Λ is isometric to BCC. Consequently, for $\nu > 2$ minimisers of T_ν exist and converge to BCC as $\nu \rightarrow \infty$, with $(\min T_\nu)^{1/\nu} \rightarrow 2/3$.*

Let $G = B^T B$ be the Gram matrix of a basis, normalised by $G_{11} = \lambda_1^2 = 1$, and $Q(G) = P(\Lambda)/\sqrt{\det G}$ (scale invariant). The remaining entries $q = (g_{22}, g_{33}, g_{12}, g_{13}, g_{23})$ are the parameters.

Lemma 5. *Every lattice has, up to isometry and scaling, a Gram matrix in the region $\mathcal{R}$ given by $1 \leq g_{22} \leq g_{33}$, $0 \leq g_{12}, g_{13} \leq \frac{1}{2}$, $|g_{23}| \leq g_{22}/2$ and $1 + g_{22} + 2(\varepsilon_1 \varepsilon_2 g_{12} +$*

$\varepsilon_1 g_{13} + \varepsilon_2 g_{23}) \geq 0$ for $\varepsilon_i = \pm 1$. On $\mathcal{R}$, $\det G \geq g_{22}g_{33}/4$; hence $Q < 3/2$ whenever $g_{22}g_{33} > 64/9$.

Proof. Take a Minkowski-reduced basis and flip the signs of b_2, b_3 . The triangle $(0, b_1, b_2)$ is acute, so the covering radius of $L_2 = \mathbb{Z}b_1 + \mathbb{Z}b_2$ is its circumradius ρ , with $\rho^2 = g_{22}|b_2 - b_1|^2/(4(g_{22} - g_{12}^2)) \leq (1 + g_{22})/3$. Minkowski reduction places the projection of b_3 onto the plane of L_2 in its Voronoi cell, so the height h of b_3 satisfies $h^2 \geq g_{33} - \rho^2 \geq g_{33}/3$, and $\det G = (g_{22} - g_{12}^2)h^2 \geq g_{22}g_{33}/4$. The collinear triple $(0, b_1, 2b_1)$ gives $Q \leq 2/\sqrt{\det G}$. $\square$

Within $\mathcal{R}$, BCC has the single representative $q_0 = (1, 1, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3})$, found by enumerating unimodular changes of basis.

Lemma 6 (local lemma). *For a triangle $t = (0, m, n)$ put $f_t(q) = \sqrt{\ell_m \ell_n \ell_{m-n}}/\det G$ with $\ell_v = v^T G v$. At q_0 exactly 36 triangles are active, $f_t(q_0) = 3/2$, and their gradients take 6 distinct values whose convex hull contains the origin at distance c from its boundary, where $c^2 = 27/1400$ exactly ($c = 0.138873 \dots$; computed in rational arithmetic). On $q_0 + [-r, r]^5$ with $r = 0.007$, interval arithmetic gives $\|\nabla^2 f_t\|_F \leq M = 14.92$ for all active t . Hence $\min_t f_t(q_0 + \delta) \leq \frac{3}{2} - c|\delta| + \frac{M}{2}|\delta|^2 < \frac{3}{2}$ for $0 < |\delta| \leq \sqrt{5}r = 0.0157 < 2c/M = 0.0186$.*

Branch and bound. On a box B of parameters each ℓ_v is linear in q and $\det G$ is a cubic polynomial, so $Q \leq \min_{t \in S} (\prod \max_B \ell)^{1/2} (\min_B \det G)^{-1/2}$ for any set S of candidate triangles; we use all $(0, m, n)$ with $m, n \in \{-1, 0, 1\}^3$ and the collinear $(0, m, 2m)$. Starting from the compact box $[1, \frac{8}{3}] \times [1, \frac{64}{9}] \times [0, \frac{1}{2}]^2 \times [-\frac{4}{3}, \frac{4}{3}]$, boxes are discarded when they violate a constraint of $\mathcal{R}$, lie in $q_0 + [-r, r]^5$, or have upper bound $< 3/2$, and bisected otherwise. All bounds are evaluated with directed (outward) rounding, so that each computed upper bound is a true upper bound (Appendix B). The run terminates after 9868069 boxes with every box certified. With Lemmas 5 and 6 this proves $Q \leq 3/2$ on $\mathcal{R}$ with equality only at q_0 . The convergence statement follows as in Theorem 2.

5 Exact minimisers for large exponents

Theorems 2 and 3 identify the *limits* of the minimisers. We now show that the max-min structure behind them determines the minimisers themselves for all large ν : exactly in

$d = 3$, and up to an exponentially small error in $d = 2$. The mechanism is a local statement about smoothed max–min problems. It extends to many-body energies the classical argument of Ryschkov for the Epstein zeta function at large exponents [27].

Let $U = B_{\rho_0}(0) \subset \mathbb{R}^n$ and $T_v(\delta) = \sum_{t \in I} e^{-v h_t(\delta)}$ for a countable family of C^2 functions h_t on U . Suppose that a finite *active set* $A \subset I$ has $h_t(0) = 0$ and:

- (H1) $\|\nabla^2 h_t\| \leq M$ on U for $t \in A$;
- (H2) $\sum_{t \notin A} (1 + |\nabla h_t| + |\nabla h_t|^2 + \|\nabla^2 h_t\|) e^{-v h_t} \leq C_0 e^{-v\gamma}$ on U for some $\gamma > 0$ and all $v \geq v_1$;
- (H3) 0 is an interior point of $\text{conv}\{a_t : t \in A\}$, $a_t = \nabla h_t(0)$; equivalently $c := \min_{|V|=1} \max_{t \in A} (-a_t \cdot V) > 0$.

Condition (H3) says that $\min_t h_t$ has a strict, conical local maximum at 0. It is the analogue of Voronoi's perfection and eutaxy.

Lemma 7. *The function $\Phi(\zeta) = \sum_{t \in A} e^{-a_t \cdot \zeta}$ is strictly convex and coercive, with a unique minimiser ζ^* . There are $\rho > 0$, v_0 and C such that for all $v \geq v_0$, T_v has exactly one critical point δ_v in B_ρ . It is the unique minimiser of T_v on $\overline{B_\rho}$, it is non-degenerate, and $|v\delta_v - \zeta^*| \leq C/v$.*

Proof. By (H3), $\Phi(\zeta) \geq e^{c|\zeta|}$, and the a_t span $\mathbb{R}^n$, so $\nabla^2 \Phi = \sum e^{-a_t \cdot \zeta} a_t a_t^\top$ is positive definite. Choose $\rho \leq \rho_0$ with $M\rho \leq c/4$, and let v be so large that $K_1/v \leq \rho$, with K, K_1 defined below.

Annulus. Let $\delta = sV$ with $|V| = 1$ and $0 < s \leq \rho$. Let $m = \min_A a_t \cdot V \leq -c$ and $E = e^{-v(sm + Ms^2/2)}$. Put $A_1 = \{t : a_t \cdot V \leq m + c/2\}$. For $t \in A_1$, $\nabla h_t(\delta) \cdot V \leq -c/4$, and the minimising index has weight $e^{-v h_t} \geq E$. The terms in $A \setminus A_1$ have weight at most $E e^{-vsc/4}$. Since $sm + Ms^2/2 \leq -7cs/8$, we have $E \geq 1$, so the inactive terms are at most $C_0 e^{-v\gamma} E$. Hence $\sum_I e^{-v h_t} \nabla h_t \cdot V \leq -cE/8 < 0$, i.e. $\partial_s T_v(sV) > 0$, as soon as $vs \geq K := \max\{0, \frac{4}{c} \log \frac{16N(\max\{|a_t| + M\rho\})}{c}\}$ and $C_0 e^{-v\gamma} \leq c/16$; here $N = |A|$. All bounds are uniform in V .

Inner ball. On $|\zeta| \leq K_1 := \max(K, |\zeta^*| + 1)$, the rescaled function $\Psi_v(\zeta) = T_v(\zeta/v)$ satisfies $\|\nabla^k \Psi_v - \nabla^k \Phi\| \leq C_k/v$ for $k = 1, 2$, by Taylor expansion of the active h_t and by (H2). Since $\nabla^2 \Phi \geq \lambda > 0$ there, Ψ_v is $\lambda/2$ -strongly convex on B_{K_1} for large v , so it has at most one critical point there. As $|\nabla \Psi_v(\zeta^*)| \leq C_1/v$, strong convexity puts the minimiser of Ψ_v over $\overline{B}(\zeta^*, 4C_1/(\lambda v)) \subset B_{K_1}$ in the interior, so it is a critical point, within $O(1/v)$ of ζ^* .

Every critical point in B_ρ lies in $B_{K/v}$ by the first step. The minimum over $\overline{B_\rho}$ is not attained on the boundary, so it is attained at the unique critical point. $\square$

Corollary 1. *Let $E_v = C_v \sum_I e^{-v h_t}$, $C_v > 0$, be a lattice energy. Suppose that minimisers exist and that every choice of minimisers converges to a lattice L_0 as $v \rightarrow \infty$. Suppose also that (H1)–(H3) hold in an orbifold chart at L_0 . (Any lift of a minimiser to the chart is a local minimiser of the chart function, hence a critical point.) Then for all large v the minimiser is unique and equals δ_v . If moreover L_0 is a critical point of every E_v , then it is the minimiser.*

Theorem 4. *There is v_0 such that for every $v \geq v_0$ the body-centred cubic lattice is the unique minimiser of T_v among all lattices in $\mathbb{R}^3$ of unit covolume.*

Proof. We use $h_t = \log(\pi_t/\frac{3}{2})$ for ordered pairs $t = (x, y)$, in the Gram chart of Sect. 4. Convergence holds by Theorem 3. The active set consists of the 72 ordered pairs of the 36 triangles of Lemma 6. (H3) holds with $c \geq \frac{2}{3}\sqrt{27/1400}$ by that lemma, since $\nabla h_t = \nabla f_t/f_t$. (H1) holds because the h_t are smooth near q_0 and A is finite. For (H2): the derivatives of $\log \ell_v$ in the chart are bounded uniformly in v on a small neighbourhood U . Only finitely many triangles have product below any bound, and the 36 triangles of Lemma 6 are the only ones with product $\frac{3}{2}$ (the next value is $\frac{3}{2}\sqrt{2}$). By continuity, $\pi_t \geq \frac{3}{2}e^{2\gamma}$ on U for all inactive t , for some $\gamma > 0$. Then the inactive sum is at most $e^{-(v-v_1)2\gamma} \sup_U \sum(\dots) e^{-v_1 h_t}$ for any $v_1 > 2$. (H3) holds with $c = \frac{2}{3}\sqrt{27/1400}$, by Lemma 6 and $\nabla h_t = \nabla f_t/f_t$. Finally, BCC is critical for every v . On the full space of Gram matrices, T_v is invariant under scaling and under the point group O_h of BCC, which acts on trace-free quadratic forms as $E_g \oplus T_{2g}$, without trivial summand. So the full gradient at BCC vanishes, and hence so does its restriction to the slice $g_{11} = 1$. $\square$

Theorem 5. *There are v_0 and $c_0 > 0$ such that for all $v \geq v_0$ the unique minimiser of T_v in $d = 2$ is the rectangular lattice Λ_{iy_v} . Here $|y_v - \hat{y}_v| \leq e^{-c_0 v}$, where $\hat{y}_v$ is the root near y_∞ of*

$$\left(\frac{y\sqrt{1+y^2}}{2}\right)^v = \frac{4(y^2-1)}{3(y^2+1)}. \quad (4)$$

In particular $y_v = y_\infty - \kappa/v + O(v^{-2})$ with

$$\kappa = \frac{y_\infty(y_\infty^2+1)}{2y_\infty^2+1} \log \frac{3(y_\infty^2+1)}{4(y_\infty^2-1)} = 0.9548949636\dots,$$

and the minimiser of $C_{s,s,s}$ is $\tau_s = iy_{2s}$ for all large s .

Proof. Work in the chart $\tau = x + iy$ at iy_∞ ; convergence holds by Theorem 2. By Lemma 4 and its proof, the active set consists of two families. The first is the collinear triples along u : 6 ordered pairs with $\pi = 2y^{-3/2}$. The second is the 12 right triangles of the four rectangles at 0: 24 ordered pairs; for $(0, 1, \tau)$, $\pi^2 = |\tau|^2|\tau-1|^2/y^3$. At iy_∞ their gradients are $(0, -\beta)$ (6 pairs), (α, γ) and $(-\alpha, \gamma)$ (12 pairs each), with $\beta = \frac{3}{2y_\infty}$, $\alpha = \frac{1}{1+y_\infty^2}$ and $\gamma = \frac{y_\infty^2-1}{2y_\infty(1+y_\infty^2)}$, all positive. So (H3) holds; (H1) and (H2) follow as in the proof of Theorem 4, with a gap $\gamma' > 0$ for all inactive terms, uniformly near iy_∞ . Lemma 7 gives a unique critical point near iy_∞ ; since $T_v(-\bar{\tau}) = T_v(\tau)$, it lies on the imaginary axis. On the axis the active terms in each family coincide, and all other terms are smaller by a factor $e^{-v\gamma'}$. Hence $\partial_y T_v = 0$ reads

$$6H_1' e^{-vH_1} + 24H_2' e^{-vH_2} = O(e^{-v\gamma'}) e^{-vH_1},$$

with $H_1 = \log(2y^{-3/2})$ and $H_2 = \frac{1}{2} \log(y + y^{-1})$. Since $-4H_2'/H_1' = \frac{4(y^2-1)}{3(y^2+1)}$ and $H_2 - H_1 = \log \frac{y\sqrt{1+y^2}}{2}$, this is (4) up to an exponentially small perturbation. The y -derivative of the equation is of order v , which gives the bound on $y_v - \hat{y}_v$. Linearising at y_∞ , where $y\sqrt{1+y^2} = 2$, gives κ ; it is also $-\zeta_y^*$ for $\Phi(\zeta) = 6e^{\beta\zeta_y} + 12e^{-\alpha\zeta_x - \gamma\zeta_y} + 12e^{\alpha\zeta_x - \gamma\zeta_y}$. $\square$

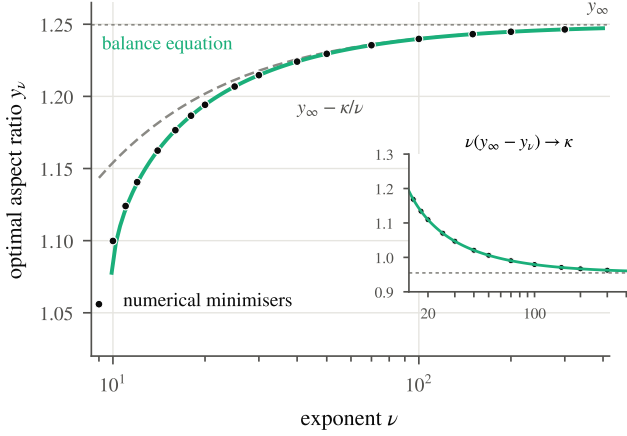


Fig. 2 Optimal aspect ratio y_v in the rectangular regime: numerical minimisers (dots), the root of the balance equation (4) (line) and the first-order law $y_\infty - \kappa/\nu$ (dashed). Inset: $\nu(y_\infty - y_v) \rightarrow \kappa$.

Numerically, the root of (4) agrees with the minimiser computed by direct summation to $6 \cdot 10^{-7}$ at $\nu = 20$ and to better than 10^{-9} for $\nu \geq 50$. A fit gives $y_v = y_\infty - 0.95489/\nu - 2.319/\nu^2 + \dots$ (Fig. 2). Neither theorem gives an explicit ν_0 ; the constants produced by the proof are astronomically large. For finite exponents the rectangular regime is covered rigorously at individual exponents in Sect. 6.

Remark 1. Lemma 7 and Corollary 1 also apply to the lattice theta function $\theta(L, \alpha) = \sum_{v \in L} e^{-\pi \alpha |v|^2}$ as $\alpha \rightarrow \infty$. One minimises $\theta - 1$, i.e. sums over $v \neq 0$, with $v = \pi \alpha$ and $h_v = |v|^2 - \lambda_1(L_0)^2$ at unit covolume. Minimisers converge to the densest lattice, by $\theta - 1 \geq 2e^{-\pi \alpha \lambda_1^2}$, Mahler compactness and uniqueness of the densest lattice. Condition (H3) becomes Voronoi's condition that L_0 is perfect and strictly eutactic: perfection gives the spanning, and positive eutaxy coefficients put 0 in the interior. In dimensions $4 \leq d \leq 7$ the densest lattices D_4, D_5, E_6, E_7 are unique (Korkine–Zolotarev, Blichfeldt, Vetčinkin), perfect and eutactic. Their automorphism groups act absolutely irreducibly, which makes them critical for every α . So each is the unique minimiser of $\theta(\cdot, \alpha)$ among lattices of its dimension for all sufficiently large α , and by Poisson summation its dual is the unique minimiser for all sufficiently small α . The Epstein analogue is due to Ryshkov [27], and local versions are due to Sarnak–Strömbergsson and Coulangeon–Schürmann [28, 29]. We did not find the global theta statement in the literature, but it may well be known to experts.

6 Finite exponents: computer-assisted proofs

Theorem 6 (computer-assisted). *Among planar lattices of unit covolume, the square lattice is the unique minimiser of T_v for $v \in \{4, 5, 6, 7, 8\}$, and the hexagonal lattice is the unique minimiser for $v = 3.5$. In particular $C_{2,2,2}$, $C_{3,3,3}$ and $C_{4,4,4}$ attain their minimum on the fundamental domain only at $\tau = i$. Moreover $T_v(\mathbb{Z}^2) - T_v(\Lambda_\rho)$ changes sign in $(3.918364, 3.918366)$, and the second derivative of $y \mapsto T_v(\Lambda_{iy})$ at $y = 1$ changes sign in $(8.604, 8.608)$; in particular the square lattice is not a local minimiser for $v = 8.608$.*

Table 1 Data of the rigorous computer-assisted proofs of Theorem 6.

v	point	$T_v(\tau_0)$ (enclosure)	$c_{20} \geq$	r_{loc}	boxes
3.5	ρ	[9.77228723809, 9.77246454156]	2.127	0.0018	373
4	i	[7.58266977279, 7.58267002399]	0.4368	0.0031	531
5	i	[4.83802684659, 4.83802684684]	2.5821	0.0069	303
6	i	[3.24997431242, 3.24997431246]	4.2719	0.0076	307
7	i	[2.23344617477, 2.23344617480]	5.3445	0.0050	323
8	i	[1.55240121256, 1.55240121259]	5.8308	0.0026	385

Together with the case $s = 1$ and Theorem 2, the minimiser of $C_{s,s,s}$ is thus ρ for $s = 1$, i for $s = 2, 3, 4$, and tends to iy_∞ as $s \rightarrow \infty$.

A universal majorant. For $v = (m, n)$ and $\tau = x + iy$ let $\ell_v = |m + n\tau|^2/y$, so that $T_v = \sum \prod_{\text{sides}} \ell_v^{-s}$ with $s = v/2$. Around (x_0, y_0) put $A = a/y_0$, $B = b/y_0$. A short computation gives

$$\frac{\ell_v(x_0 + a, y_0 + b)}{\ell_v(x_0, y_0)} = 1 + \frac{(2\alpha^2 - 1)B + 2\alpha\beta A + \alpha^2(A^2 + B^2)}{1 + B},$$

where $\alpha^2 + \beta^2 = 1$ depend on v . Coefficientwise, therefore, $\ell/\ell_0 - 1 \ll \hat{\epsilon} := (A + B + A^2 + B^2)/(1 - B)$, and every triangle term t satisfies $t/t_0 \ll (1 - \hat{\epsilon})^{-3s}$, uniformly in the lattice vectors. Consequently all Taylor coefficients of T_v at τ_0 are bounded by $T_v(\tau_0)$ times the coefficients $\hat{G}_{ij}$ of $(1 - \hat{\epsilon})^{-3s}$. This yields explicit Taylor remainders and explicit bounds for the terms omitted by a finite truncation.

Proof structure. Let $\tau_0 \in \{i, \rho\}$ be the candidate. (i) For $y > Y_0$ the collinear triples along the vector 1 give $T_v(\tau) \geq y^{3s} \sum_{j \neq k} (|j||k||j - k|)^{-2s} > T_v(\tau_0)$. (ii) Local lemma: the reflections fixing τ_0 ($x \mapsto -x$ and $\tau \mapsto 1/\bar{\tau}$ at i ; $x \mapsto 1 - x$ and the order-three rotation at ρ) force $c_{10} = c_{01} = c_{11} = c_{30} = c_{12} = 0$ in the Taylor expansion. With rigorous enclosures of $c_{20}, c_{02}, c_{21}, c_{03}$ and the majorant remainder, $T(\tau_0 + \delta) - T(\tau_0) \geq A^2(c_{20} - |c_{21}|r - \rho) + B^2(c_{02} - |c_{03}|r - \rho) > 0$ for $0 < \max(|A|, |B|) \leq r_{\text{loc}}$, where $\rho = T \sum_{i+j \geq 4} \hat{G}_{ij} r^{i+j-2}$. (iii) The rest of $[0, \frac{1}{2}] \times [0.8, Y_0]$ is covered by boxes on which the second-order Taylor polynomial at the centre, with rigorous coefficient enclosures, minus the majorant remainder exceeds an upper bound for $T(\tau_0)$. All coefficients are enclosed rigorously: the summands are evaluated in outward-rounded interval arithmetic (powers in Arb [25]), the cube-truncated sums ($R = 40$) by direct, non-FFT convolution with a Higham-type bound for the accumulated rounding error, and the omitted terms by an explicit tail bound (Appendix B). All comparisons are done on the resulting enclosures. Table 1 lists the data.

The crossing statement uses the same enclosures at $R = 100$ and gives $T_v(\mathbb{Z}^2) - T_v(\Lambda_\rho) \in [2.31, 2.90] \cdot 10^{-7}$ at $v = 3.918364$ and $\in [-3.46, -2.87] \cdot 10^{-7}$ at $v = 3.918366$. For v_2 , the coefficient c_{02} at $\tau = i$, i.e. half the second B -derivative, is enclosed in $[1.1544, 1.1545] \cdot 10^{-3}$ at $v = 8.604$ and in $[-1.3452, -1.3451] \cdot 10^{-3}$ at $v = 8.608$, while $c_{20} \approx 5.89 > 0$ at both exponents.

Remark 2 (status of the computer-assisted parts). All computer-assisted statements (Theorems 3 and 6) are proved with rigorous error control. Every floating-point operation that enters a certificate is either rounded outward (interval

arithmetic with `nextafter`, relying on correctly rounded IEEE-754 $+$, $-$, $\times$, $\div$, $\sqrt{}$ or evaluated in the ball arithmetic of Arb [25]; large sums are computed by direct convolution in double precision with an a-priori bound of the accumulated rounding error of the form $\gamma_n \sum |a_i| |b_j|$, $\gamma_n = nu/(1 - nu)$ [26]. The trusted base therefore consists of IEEE-754 double arithmetic, the direct convolution routine of SciPy, Arb (via python-flint), and the short Python drivers. The code, the logs and the certificates are publicly available (see Declarations). As independent sanity checks, the three-dimensional box bounds were compared with brute-force values at 1200 random points, and the provers fail, as they must, when asked to certify false statements such as $Q < 1.49$.

The same method did not succeed for the hexagonal lattice at $v = 3$. Close to $v = d$ the truncation tail decays only like R^{2-2v} with a large constant, and at $R = 40$ the error exceeds the energy gap next to ρ .

Continua of exponents. Theorem 6 covers individual exponents. Extending it to an interval such as $v \in [4, 8]$ requires enclosures that are uniform in v ; the majorant is monotone in s , and ∂_v of every summand is bounded by log of the side-length product times the summand, so a covering of $[4, 8]$ by v -intervals is possible in principle. With the present implementation, where one exponent costs 20–40 minutes, a covering fine enough near $v = 4$ (where c_{20} is small) would need of the order of 10^2 exponent intervals and several CPU days, so we leave it for future work.

7 Numerical phase diagram

The numerical results use an independent reference summation, described in Appendix A. It agrees with the library of [13] to better than $2 \cdot 10^{-10}$ relative in 55 test cases and reproduces the published values of [11] at $v = 3$.

Two dimensions. A global search over the fundamental domain gives the following picture (Figs. 3–5).

- For $2.2 \leq v < v^*$ the global search returns the hexagonal lattice. Below $v = 2$ we only compared the four candidates hexagonal, square, rectangular ($y = 1.1$) and rhombic (75°) at $v = 1.4, 1.5, \dots, 1.9$, using an extrapolation ansatz with the two exponent families R^{d-2v} and R^{2d-3v} ; the hexagonal lattice is the best of them, with a stability of 10^{-8} across cut-off sets. The range $4/3 < v < 1.4$ was not computed.
- At $v^* = 3.9183649026$ the global minimiser switches from the hexagonal to the square lattice. The switch is discontinuous: both lattices are local minima for $3.80632 < v < 4.27848$ (errors below $6 \cdot 10^{-5}$), their energies simply cross, and along the arc $|\tau| = 1$ the barrier between them is 0.14%.
- At $v_2 = 8.6063 \pm 0.0005$ (rigorously $v_2 \in (8.604, 8.608)$, Theorem 6) the square lattice becomes unstable towards rectangular deformations, and the optimal aspect ratio then grows continuously towards y_∞ .

Error bars are the spread over three finite-difference steps and two cut-off sets. We use the words *switch* and *bifurcation*

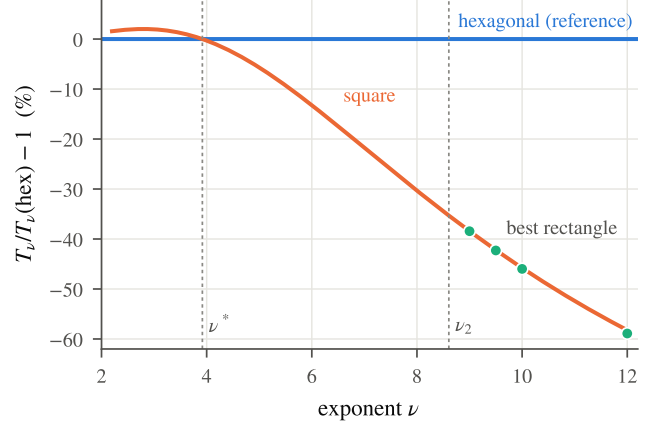


Fig. 3 Relative energy of the square lattice (line) and of the best rectangular lattice (dots) with respect to the hexagonal lattice. The zeros mark the switch at v^* and the bifurcation at v_2 .

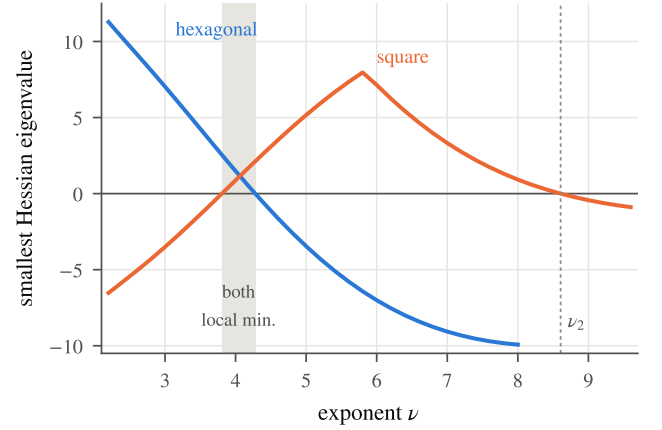


Fig. 4 Smallest Hessian eigenvalue of T_v at the hexagonal and at the square point. Grey: both are local minima, so the switch at v^* is discontinuous. Right: loss of stability of the square lattice at v_2 .

in the sense of the lattice-minimisation literature: a change of the global minimiser as the parameter v varies. At zero temperature this is a level crossing of two local minima (at v^*) or a loss of stability (at v_2); no thermodynamic phase transition is implied.

Conjecture 1. In $d = 2$ the minimiser of T_v is hexagonal for $4/3 < v < v^*$, square for $v^* < v \leq v_2$, and rectangular with aspect ratio increasing from 1 to y_∞ for $v > v_2$.

Unequal exponents. Replacing the exponent of the side $|x - y|$ by μ gives, up to normalisation, $C_{v/2, v/2, \mu/2}$. The curve of Fig. 1(b) then becomes a phase map (Fig. 6). The square phase occupies a broad band around the diagonal; for $4 \leq v \leq 5.5$ the hexagonal lattice returns at large μ ; the rectangular phase appears only for $v \gtrsim 9$ in a window $8.5 \lesssim \mu \lesssim 11$. No rhombic or oblique minimiser was found.

Three dimensions. A global search over all Bravais lattices used 67 starts per exponent (FCC, BCC, SC and 64 random), Nelder–Mead and accurate re-evaluation. It returned BCC as the best lattice, to 10^{-9} relative, for every tested $v \in \{3.2, 3.5, 4, 4.5, 5, 6, 7, 8, 12\}$; between 87% and 100% of the starts converged to BCC. For $2.2 \leq v \leq 2.8$ we only

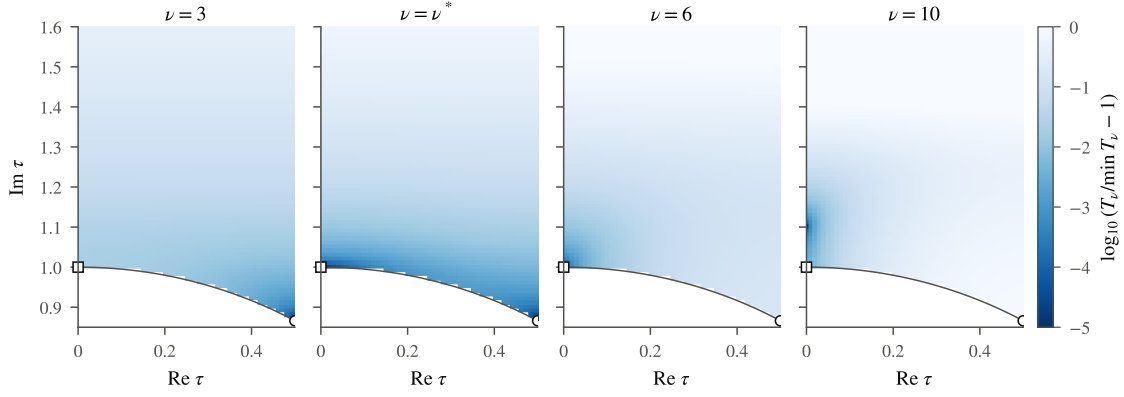


Fig. 5 Energy landscape $\log_{10}(T_\nu/\min T_\nu - 1)$ on the fundamental domain for four exponents. Circle: hexagonal point; square: square point. At $\nu = 10$ the minimum lies on the imaginary axis above $\tau = i$.

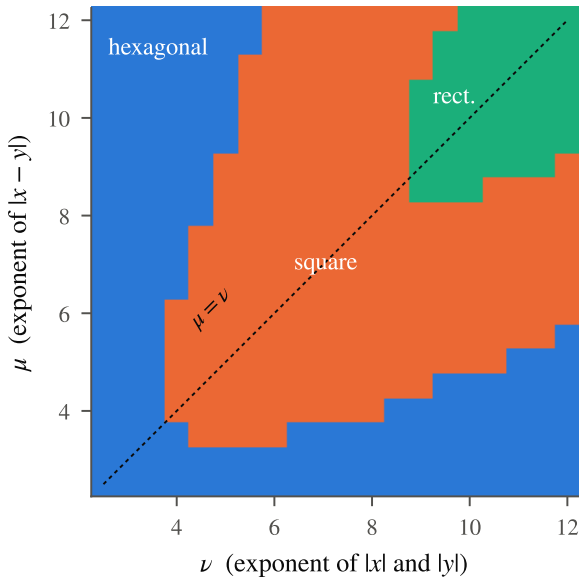


Fig. 6 Minimiser for the exponents (ν, ν, μ) on a grid of spacing 0.5. Dashed: equal exponents.

compared BCC, FCC and SC; the ordering $\text{BCC} < \text{FCC} < \text{SC}$ persists. The curvature of T_ν at FCC along the Bain path changes sign at $\nu_c = 3.7521 \pm 0.0001$, so FCC is a saddle point for larger ν . Optimised over $c/a \in [1.45, 1.85]$, hcp lies slightly above FCC for all tested $\nu \in \{3.5, 4.5, 6, 9\}$ and is therefore never optimal there; other non-Bravais structures were not examined; at $\nu = 4.5$, for example, the values are 17.9724 against 17.9624, while BCC gives 17.7406. BCC is stationary along the Bain path for every exponent [11].

8 Mixed pair and three-body energies

Physical three-body terms always come with pair terms. As a first step we consider $E_\lambda = Z_\nu + \lambda T_\nu$ at unit density, where $Z_\nu = \sum' |x|^{-\nu}$ is the Epstein (Riesz) pair energy, evaluated with [14], and $\lambda = c_3/c_2 \geq 0$ is the ratio of the coupling constants. Both terms use the same exponent. Pair repulsion favours the hexagonal lattice in $d = 2$ and, numerically, FCC in $d = 3$; the three-body term favours the square lattice (for $\nu > \nu^*$) and BCC, respectively. For two competing lattices L_1 (preferred by Z_ν) and L_2 (preferred by T_ν) the energies

cross at

$$\lambda_c = \frac{Z_\nu(L_2) - Z_\nu(L_1)}{T_\nu(L_1) - T_\nu(L_2)} = \frac{Z_\nu(L_1)}{T_\nu(L_1)} \cdot \frac{\delta Z}{\delta T}, \quad (5)$$

where δZ and δT are the relative energy gaps. We computed λ_c on a grid of spacing 0.25 in ν (Fig. 7). In $d = 3$ we compared FCC, BCC, hcp with optimised $c/a \in [1.45, 1.85]$ and the Bain family. In $d = 2$ we compared the hexagonal, the square and the rectangular lattices.

Three dimensions. The minimiser jumps directly from FCC to BCC, and hcp was never optimal among the compared structures. The critical ratio grows monotonically, from $\lambda_c = 0.049$ at $\nu = 3.5$ through 0.103 at $\nu = 6$ and 0.289 at $\nu = 9$ to 0.81 at $\nu = 12$. The pair energy distinguishes FCC and BCC only weakly: the relative gap δZ is $2 \cdot 10^{-4}$ at $\nu = 3.5$ and 5.5% at $\nu = 12$. The three-body gap δT is larger by a factor between 5 and 15. The growth of λ_c therefore comes mainly from the prefactor Z_ν/T_ν in (5). This prefactor grows exponentially. For large ν one has $Z_\nu \approx k_2 \lambda_1^{-\nu}$ and $T_\nu \approx k_3 P^{-\nu}$, with λ_1 the shortest distance and P the smallest triangle product. Hence Z_ν/T_ν grows like $(P/\lambda_1)^\nu = \lambda_1^{2\nu}$ for FCC ($P = \lambda_1^3$), i.e. by a factor of about 2 per three units of ν , as observed.

Two dimensions. The minimiser jumps from hexagonal to square, and λ_c is *not* monotone. It diverges as $\nu \downarrow \nu^*$, has a minimum $\lambda_c \approx 1.04$ at $\nu \approx 6.25$, and grows again, to $\lambda_c \approx 2.9$ at $\nu = 12$. Both effects follow from (5). Near ν^* the three-body advantage of the square lattice vanishes, $\delta T \rightarrow 0$, while the pair penalty $\delta Z \approx 4\%$ stays finite. For large ν the relative gaps δZ and δT become comparable (0.60 and 0.58 at $\nu = 12$), and the prefactor $Z_\nu/T_\nu \sim \lambda_1^{2\nu}$ (with $\lambda_1 = 1.0746$ for the hexagonal lattice) takes over. For $\nu < \nu^*$ both terms favour the hexagonal lattice and there is no switch. At $\nu = 9$, with $\nu > \nu_2$, a further switch to a rectangular lattice occurs at very large λ .

The numerical values of λ_c depend on the normalisation of the two terms and should be read as a comparison between exponents, not as physical coupling constants. For the ATM energy the three-body term carries an angular factor that changes sign, so these results do not transfer directly.

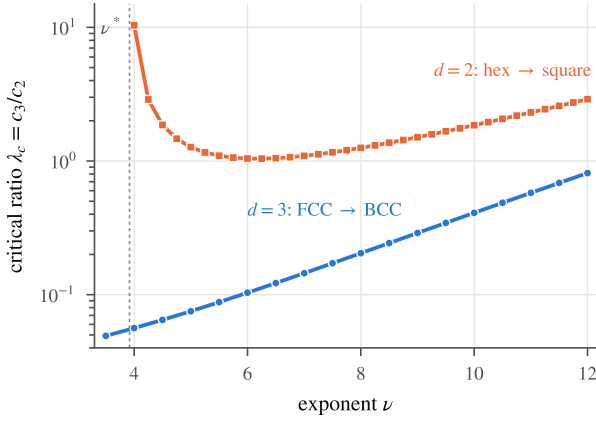


Fig. 7 Critical ratio $\lambda_c = c_3/c_2$ of the mixed energy $Z_\nu + \lambda T_\nu$, above which the three-body favourite wins: BCC over FCC in $d = 3$, square over hexagonal in $d = 2$. Dashed: the crossing ν^* of the pure three-body energy.

9 Discussion

A purely three-body repulsion of product type selects different lattices than pair repulsion does. In the plane it penalises the equilateral triangles of the hexagonal lattice once the decay is steep enough. In space it favours BCC, whose smallest triangles are isosceles with one second-neighbour side, over the close-packed FCC and hcp. The steep-decay limits reduce to max–min problems for triangles. The planar one has the closed-form answer of Theorem 1; the spatial one we solved by a computer-assisted argument. Through (3) the planar results translate into statements on the minima of the modular graph functions $C_{s,s,s}$.

Several questions remain open. (i) A proof of the full planar phase diagram, in particular hexagonal optimality for all $\nu < \nu^*$, would require the finite- ν method on continua of exponents (Sect. 6) and sharper tail bounds near $\nu = d$, for instance via the Epstein-zeta representation of [12]. (ii) An analytic proof of Theorem 3 is missing, and so are explicit values of ν_0 in Theorems 4 and 5; the latter would combine with the finite- ν proofs into a complete description of the steep-decay regime. (iii) Physically relevant three-body energies include the angular ATM factor; the mixed energies of Sect. 8 are only a first step. (iv) Theorem 5 and the finite- ν proofs locate the minimiser on the imaginary axis for large ν and at individual exponents; that this holds for every $\nu > \nu_2$ is supported numerically but unproven.

Declarations

Affiliation The author is a student at ETH Zürich; this work was carried out independently of any course or research group.

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Code and data availability All scripts, raw data, logs and certificates are publicly archived at Zenodo, <https://doi.org/10.5281/zenodo.22975868>.

This includes the rigorous provers (rigorous/), the Lean formalisation (lean/) and the numerical experiments (extended/). Each prover is a single Python script that prints its certificate and can be re-run on a laptop. The planar provers take 20–40 minutes per exponent, and the three-dimensional branch and bound takes about 4 minutes.

Competing interests The author declares no competing interests.

A Numerical methods and error control

Reference summation. T_ν is summed over all integer labels in the cube $[-R, R]^d$ for the two free vertices, with the kernel of the third side on $[-2R, 2R]^d$. The inner convolution is evaluated by FFT for exploratory computations and by direct convolution in the rigorous parts. The truncation error is dominated by configurations with one distant vertex and decays like $R^{d-2\nu}$ for $\nu > d$. For $2d/3 < \nu \leq d$ it decays like $R^{2d-3\nu}$ (with logarithms at $\nu = d$); Richardson extrapolation uses the corresponding exponents.

Tail bound. A term is missing only if a free vertex x lies outside the cube. For fixed x , $\sum_y |y|^{-\nu} |x - y|^{-\nu} \leq 2(|x|/2)^{-\nu} Z_\nu$, and $|Am| \geq \sigma_{\min}(A)|m|_\infty$, whence $0 \leq T - T_R \leq 2^{v+2} Z_\nu 8 \sigma_{\min}^{-2\nu} R^{2-2\nu} / (2\nu - 2)$ in $d = 2$.

Non-Bravais structures. For a crystal $\bigcup_j (L + s_j)$ the per-atom energy decomposes into sums $\sum_m F_{ij}(m) (F_{ik} * G_{jk})(m)$ over the Bravais lattice L . The implementation reproduces the FCC value with a four-atom cubic basis to 10^{-10} .

Validation. Besides the agreement with [13, 11] mentioned above, the reference summation reproduces the identities $C_{1,1,1} = E_3 + \zeta(3)$ (with logarithmic extrapolation, since all exponents equal d) and $C_{2,2,1} = \frac{2}{5}E_5 + \frac{\zeta(5)}{30}$ [15] to 10^{-14} and 10^{-15} .

B The computer-assisted proofs in detail

Rigorous arithmetic. An interval is a pair of doubles $[x, \bar{x}]$. Sums, products, quotients and square roots are computed in round-to-nearest and then widened by one ulp in each direction with `nextafter`. Since IEEE-754 rounds these operations correctly, the result encloses the exact value. Non-algebraic functions (x^{-s} for non-integer s , binomial coefficients $\binom{-3s}{j}$, the majorant $(1 - \hat{\epsilon})^{-3s}$ and its Taylor coefficients) are evaluated in Arb [25] with 128-bit precision and rounded outward to doubles. A sum $\sum_i a_i b_i$ of n products, computed in floating point as $\hat{S}$ from rounded values $\hat{a}_i, \hat{b}_i$ enclosed with radii r_i, ρ_i , satisfies

$$\left| \sum a_i b_i - \hat{S} \right| \leq \gamma_n \sum |\hat{a}_i| |\hat{b}_i| + \sum ((|\hat{a}_i| + r_i)(|\hat{b}_i| + \rho_i) - |\hat{a}_i| |\hat{b}_i|),$$

with $\gamma_n = nu / (1 - nu)$ and $u = 2^{-53}$ [26]. The two sums on the right are themselves computed in floating point and inflated by the factor $1 + \gamma_n$. We apply this to each output entry of the direct convolutions, with n the total number of products; this is valid for any summation order.

Theorem 3. The parameter region $\mathcal{R}$ of Lemma 5 is intersected with $g_{22}g_{33} \leq 64/9$ and covered by the initial box

$[1, \frac{8}{3}] \times [1, \frac{64}{9}] \times [0, \frac{1}{2}]^2 \times [-\frac{4}{3}, \frac{4}{3}]$. On a box B we compute: (a) *Infeasibility*: B is discarded if a linear constraint of $\mathcal{R}$ is violated on all of B , i.e. if its maximum over the vertices, computed with upward rounding, is negative. (b) *Exclusion*: B is discarded if it lies inside $q_0 + [-0.0069, 0.0069]^5$, which is contained in the box of Lemma 6. (c) *Upper bound*: for each candidate triangle $t = (0, m, n)$, the three squared side lengths $\ell_v = v^T G v$ are linear in q , so their maxima over B are attained at vertices and are evaluated with upward rounding. For $\det G$, a cubic in q , we use the lower bound obtained by bounding each monomial from below over B with downward rounding. The bound is $U(B) = \min_t (\prod \max_B \ell)^{1/2} / (\min_B \det G)^{1/2}$, evaluated with upward rounding. The candidate triangles are the $(0, m, n)$ with $m, n \in \{-1, 0, 1\}^3 \setminus \{0\}$ and the collinear triples $(0, m, 2m)$. For each box, the 24 candidates with the smallest product at the box centre enter the minimum. Any subset gives a valid upper bound, so this heuristic choice affects only the speed. If $U(B) < 3/2$, the box is certified. (d) Otherwise B is bisected along its widest coordinate, with widths scaled by the size of the initial box. The run processed 9868069 boxes and ended with an empty queue. The local lemma (Lemma 6) is proved separately. The 36 active triangles are enumerated exactly. The minimum-norm point of the convex hull of their gradients is computed in rational arithmetic ($c^2 = 27/1400$, attained with equal weights $1/6$ on six gradients). The Hessian bound M uses interval arithmetic over the whole box $q_0 + [-r, r]^5$.

Theorem 6. Let $\tau_0 \in \{i, \rho\}$ and let (x_0, y_0) be the double nearest to τ_0 (exact for i). The proof has four steps.

1. *Jets*. For every lattice vector v in the cube $[-2R, 2R]^2$ we enclose the Taylor coefficients up to order K of ℓ_v^{-s} in $(A, B) = (a, b)/y_0$, using the closed form of $\ell_v/\ell_{v,0}$ from Sect. 6 and the binomial series. The coefficients c_{ij} of T_v are sums over triangles of products of three such jets. Each is a double sum of convolution type, evaluated by direct convolution with the error bound above, plus the tail B_R times $\hat{G}_{ij}$.
2. *Large y* . The exact lower bound $T \geq y^{3s} K_s$ with a finite partial sum K_s (Arb) gives a cut Y_0 .
3. *Local lemma*. At i , the symmetry zeros are exact, and the direct inequality of Sect. 6 is checked for $r = r_{\text{loc}}$. At ρ , where the expansion point is only within 10^{-16} of ρ , we instead bound the Hessian of T in (A, B) on the box $|A|, |B| \leq r$ from below. We use the enclosed coefficients up to order three and the majorant remainder $T \sum_{k \geq 4} g_k k^2 r^{k-2}$, where $g_k = \sum_{i+j=k} \hat{G}_{ij}$. This remainder is computed in closed form from $g(t) = (1 - h(t))^{-3s}$, $h(t) = (2t + 2t^2)/(1 - t)$, via $\sum_k g_k k^2 t^k = t g' + t^2 g''$. A positive definite Hessian on the box, together with criticality of ρ (by symmetry), shows that ρ is the unique minimiser in the box.
4. *Branch and bound* on $[0, \frac{1}{2}] \times [0.8, Y_0]$, minus the unit disc and the local box. On a box with centre τ_c and scaled half-widths $(\hat{A}, \hat{B})$, the lower bound is $c_{00} - |c_{10}| \hat{A} - |c_{01}| \hat{B} - (c_{20}) \hat{A}^2 - |c_{11}| \hat{A} \hat{B} - (c_{02}) \hat{B}^2 - T \sum_{i+j \geq 3} \hat{G}_{ij} \hat{A}^i \hat{B}^j$, all at τ_c . The box is cer-

tified if this exceeds the upper end of the enclosure of $T(\tau_0)$. Close to τ_0 , where the gap is smaller than the tail bound B_R , we compare the *truncated* sums instead. The tails at τ and at τ_0 differ by at most $B_R(\tau_0)(\hat{G}(A_0, B_0) - 1)$, where (A_0, B_0) is the scaled distance of the box from τ_0 .

For $\tau_0 = \rho$ the value used in step 4 is an upper bound of T at the double (x_0, y_0) . It is admissible because ρ minimises T on the local box, which contains (x_0, y_0) .

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